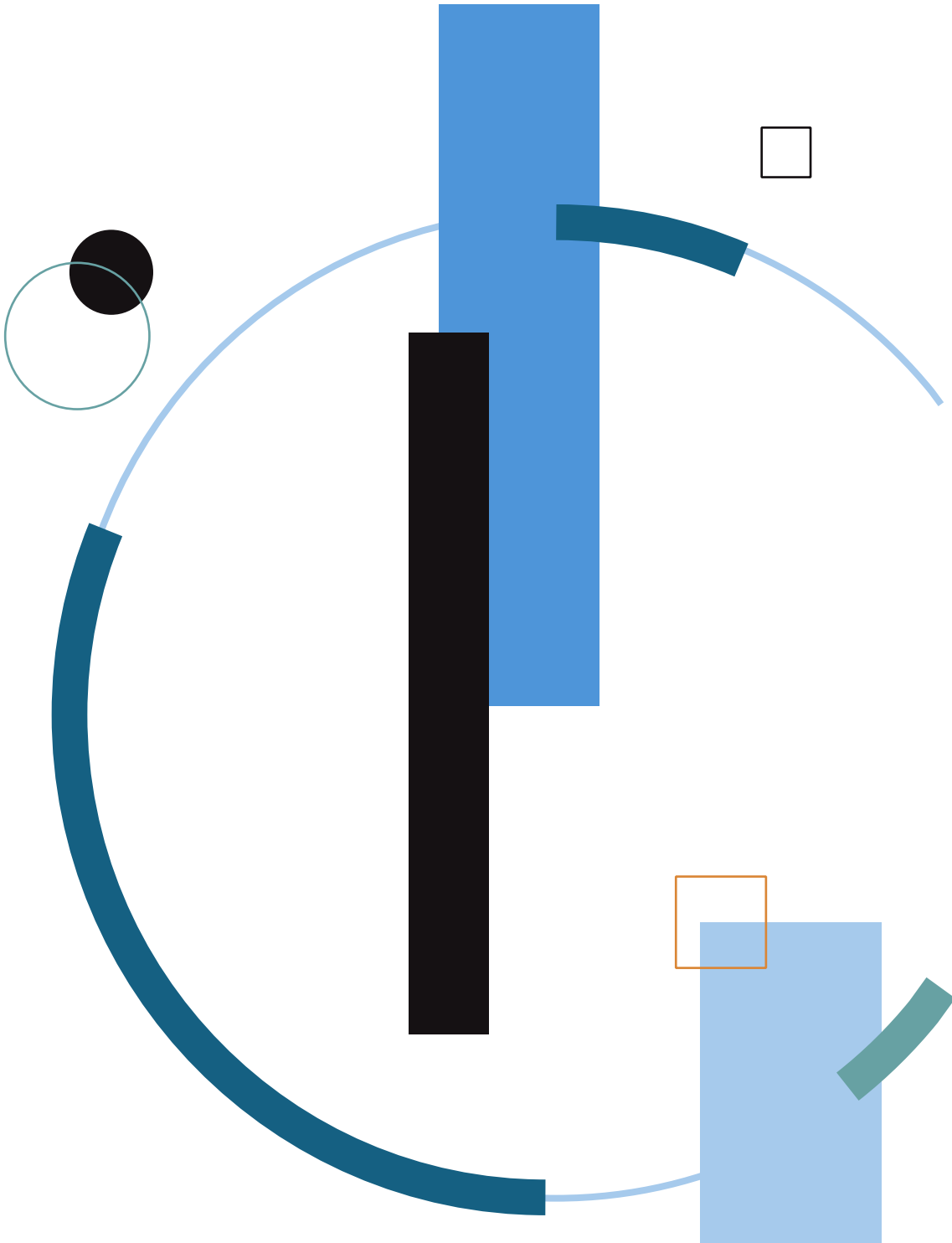


BLUE TEAM

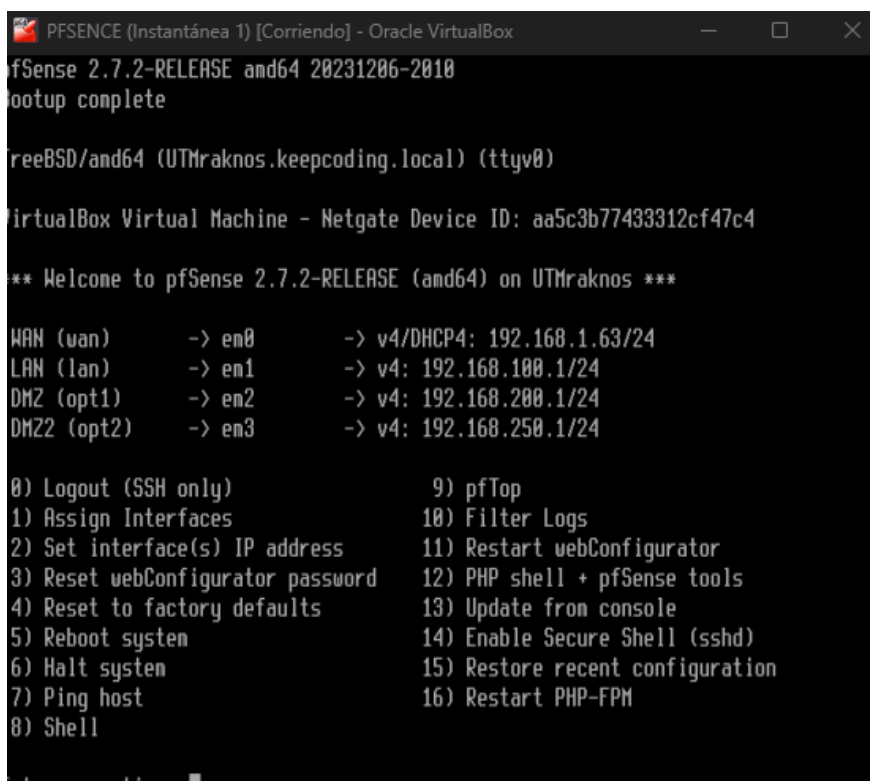
Rubén López Gálvez



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1. Infraestructura de red en pfSense.



```
PFSENSE (Instantánea 1) [Corriendo] - Oracle VirtualBox
pfSense 2.7.2-RELEASE amd64 20231206-2010
bootup complete

FreeBSD/amd64 (UTMraknos.keepcoding.local) (ttyv0)

VirtualBox Virtual Machine - Netgate Device ID: aa5c3b77433312cf47c4

** Welcome to pfSense 2.7.2-RELEASE (amd64) on UTMraknos ***

WAN (wan)      -> em0      -> v4/DHCP4: 192.168.1.63/24
LAN (lan)      -> em1      -> v4: 192.168.100.1/24
DMZ (opt1)     -> em2      -> v4: 192.168.200.1/24
DMZ2 (opt2)    -> em3      -> v4: 192.168.250.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults    13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell
```

3

Objetivo

Se ha montado una red segmentada en pfSense con tres redes internas (LAN, DMZ y DMZ2) y una red externa (WAN). El objetivo es aislar servicios por zonas y poder cumplir los requisitos del laboratorio:

- ♦ Windows en LAN enviando logs al SIEM
- ♦ Cowrie (honeypot) en DMZ
- ♦ Suricata como fuente adicional de logs en DMZ2

Topología e IPs usadas

Para garantizar la segmentación y el aislamiento de servicios, se han definido varias redes independientes gestionadas por *pfSense*. En las siguientes líneas se enumeran las interfaces, sus rangos de direccionamiento y el propósito de cada una

- ♣ WAN: 192.168.1.63 (red externa / acceso desde el host)
- ♣ LAN: 192.168.100.1/24
- ♣ DMZ: 192.168.200.1/24
- ♣ DMZ2: 192.168.250.1/24

Función de cada red:

- ♣ LAN: red de usuario (Windows)
- ♣ DMZ: red expuesta/aislada (honeypot Cowrie)
- ♣ DMZ2: red separada para sensor/fuente de logs (Suricata)
- ♣ WAN: acceso externo para pruebas controladas

Interface	Network port
WAN	em0 (08:00:27:73:31:58)
LAN	em1 (08:00:27:3b:95:3d) Delete
DMZ	em2 (08:00:27:fe:1c:f5) Delete
DMZ2	em3 (08:00:27:7c:5b:c7) Delete

Save

Interfaces that are configured as members of a lagg(4) interface will not be shown.

Wireless interfaces must be created on the Wireless tab before they can be assigned.

Configuración de interfaces en pfSense

Se han creado y asignado las interfaces LAN/DMZ/DMZ2 en pfSense, configurando en cada una su IP como puerta de enlace de la subred:

- ♥ LAN → 192.168.100.1/24
- ♥ DMZ → 192.168.200.1/24
- ♥ DMZ2 → 192.168.250.1/24

Con esto, pfSense actúa como router entre redes y permite aplicar políticas de seguridad por interfaz.

WARNING: The 'admin' account password is set to the default value. Change the password in the User Manager.

Interfaces / LAN (em1)



General Configuration

Enable	☒ Enable interface
Description	LAN Enter a description (name) for the interface here.
IPv4 Configuration Type	Static IPv4
IPv6 Configuration Type	Track Interface
MAC Address	xx:xx:xx:xx:xx:xx This field can be used to modify ("spoof") the MAC address of this interface. Enter a MAC address in the following format: xx:xx:xx:xx:xx:xx or leave blank.
MTU	 If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.
MSS	 If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.
Speed and Duplex	Default (no preference, typically autoselect) Explicitly set speed and duplex mode for this interface. WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

Static IPv4 Configuration

IPv4 Address	192.168.100.1 / 24
IPv4 Upstream gateway	None + Add a new gateway If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button. On local area network interfaces the upstream gateway should be "none". Selecting an upstream gateway causes the firewall to treat this interface as a WAN type interface . Gateways can be managed by clicking here .

Track IPv6 Interface

IPv6 Interface	WAN Selects the dynamic IPv6 WAN interface to track for configuration.
IPv6 Prefix ID	0 (hexadecimal from 0 to 0) The value in this field is the (Delegated) IPv6 prefix ID. This determines the configurable network ID based on the dynamic IPv6 connection. The default value is 0.

Reserved Networks

Block private networks and loopback addresses	☐ Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.
Block bogon networks	☐ Blocks traffic from reserved IP addresses (but not RFC 1918) or not yet assigned by IANA. Bogons are prefixes that should never appear in the Internet routing table, and so should not appear as the source address in any packets received. This option should only be used on external interfaces (WANs), it is not necessary on local interfaces and it can potentially block required local traffic. Note: The update frequency can be changed under System > Advanced, Firewall & NAT settings.

[Save](#)

Adaptador de Ethernet Ethernet:

```
Sufijo DNS específico para la conexión. . : keepcoding.local
Vínculo: dirección IPv6 local. . . : fe80::42b7:2aa9:e6a8:6132%5
Dirección IPv4. . . . . : 192.168.100.102
Máscara de subred . . . . . : 255.255.255.0
Puerta de enlace predeterminada . . . . . : fe80::a00:27ff:fe3b:953d%5
192.168.100.1
```

WARNING: The admin account password is set to the default value. Change the password in the User Manager.

Interfaces / DMZ (em2)



General Configuration

Enable ☒ Enable interface

Description

DMZ

Enter a description (name) for the interface here.

IPv4 Configuration
Type

Static IPv4

IPv6 Configuration
Type

None

MAC Address

xx:xx:xx:xx:xx:xx

This field can be used to modify ("spoof") the MAC address of this interface.
Enter a MAC address in the following format: xx:xx:xx:xx:xx:xx or leave blank.

MTU

If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS

If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex

Default (no preference, typically autoselect)

Explicitly set speed and duplex mode for this interface.

WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

Static IPv4 Configuration

IPv4 Address

192.168.200.1

/ 24

IPv4 Upstream
gateway

None

+ Add a new gateway

If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button.
On local area network interfaces the upstream gateway should be "none".

Selecting an upstream gateway causes the firewall to treat this interface as a [WAN type interface](#).

Gateways can be managed by [clicking here](#).

Reserved Networks

Block private
networks and
loopback addresses

☐

Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.

Block bogon
networks

☐

Blocks traffic from reserved IP addresses (but not RFC 1918) or not yet assigned by IANA. Bogons are prefixes that should never appear in the Internet routing table, and so should not appear as the source address in any packets received. This option should only be used on external interfaces (WANs), it is not necessary on local interfaces and it can potentially block required local traffic.

Note: The update frequency can be changed under System > Advanced, Firewall & NAT settings.

Save

```
th0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
inet 192.168.200.99 netmask 255.255.255.0 broadcast 192.168.200.255
inet6 fe80::1928:c05a:bc3d:c364 prefixlen 64 scopeid 0x20<link>
ether 08:00:27:1f:b7:23 txqueuelen 1000 (Ethernet)
RX packets 890111 bytes 1228710525 (1.1 GiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 416113 bytes 56807494 (54.1 MiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Interfaces / DMZ2 (em3)

General Configuration

Enable
☒ Enable interface

Description
DMZ2

Enter a description (name) for the interface here.

IPv4 Configuration Type
Static IPv4

IPv6 Configuration Type
None

MAC Address
xx:xx:xx:xx:xx:xx

This field can be used to modify ("spoof") the MAC address of this interface.
Enter a MAC address in the following format: xx:xx:xx:xx:xx:xx or leave blank.

MTU

If this field is blank, the adapter's default MTU will be used. This is typically 1500 bytes but can vary in some circumstances.

MSS

If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 for IPv4 (TCP/IPv4 header size) and minus 60 for IPv6 (TCP/IPv6 header size) will be in effect.

Speed and Duplex
Default (no preference, typically autoselect)

Explicitly set speed and duplex mode for this interface.
WARNING: MUST be set to autoselect (automatically negotiate speed) unless the port this interface connects to has its speed and duplex forced.

Static IPv4 Configuration

IPv4 Address
192.168.250.1
/ 24

IPv4 Upstream gateway
None

+ Add a new gateway

If this interface is an Internet connection, select an existing Gateway from the list or add a new one using the "Add" button.
On local area network interfaces the upstream gateway should be "none".
Selecting an upstream gateway causes the firewall to treat this interface as a [WAN type interface](#).
Gateways can be managed by [clicking here](#).

Reserved Networks

Block private networks and loopback addresses
☐

Blocks traffic from IP addresses that are reserved for private networks per RFC 1918 (10/8, 172.16/12, 192.168/16) and unique local addresses per RFC 4193 (fc00::/7) as well as loopback addresses (127/8). This option should generally be turned on, unless this network interface resides in such a private address space, too.

Block bogon networks
☐

Blocks traffic from reserved IP addresses (but not RFC 1918) or not yet assigned by IANA. Bogons are prefixes that should never appear in the Internet routing table, and so should not appear as the source address in any packets received.
This option should only be used on external interfaces (WANs), it is not necessary on local interfaces and it can potentially block required local traffic.
Note: the update frequency can be changed under System > Advanced, Firewall & NAT settings.

Save

```

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.250.101 netmask 255.255.255.0 broadcast 192.168.250.255
    inet6 fe80::74b7:b708:5d5d:a4aa prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:c3:42:d3 txqueuelen 1000 (Ethernet)
    RX packets 56219 bytes 51999034 (49.5 MiB)
    RX errors 0 dropped 4 overruns 0 frame 0
    TX packets 32976 bytes 12903665 (12.3 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

```



Motivo de la segmentación (LAN / DMZ / DMZ2)

La separación por redes reduce el impacto de un incidente y permite controlar y monitorizar el tráfico entre zonas aplicando políticas específicas en pfSense (reglas de firewall, NAT y control de accesos). De este modo, cada segmento tiene un propósito claro y se limita el movimiento lateral en caso de compromiso:

- * El honeypot en DMZ queda aislado para evitar que un atacante llegue a la LAN.
- * Suricata en DMZ2 está en una zona separada para generar y enviar eventos sin compartir red con el honeypot.
- * Windows permanece en LAN como equipo “normal” cuyos eventos se centralizan en el SIEM.

2. Reglas de firewall por red (WAN, LAN, DMZ y DMZ2)

El objetivo principal de estas reglas es mantener el honeypot completamente aislado de la LAN y de DMZ2, evitando movimientos laterales en caso de compromiso. Al mismo tiempo, se habilita desde WAN únicamente el acceso al servicio publicado, limitando la exposición al exterior al mínimo necesario.

► WAN (entrada desde “exterior”)

El objetivo de esta sección es habilitar el acceso externo controlado al honeypot (por ejemplo, a su servicio SSH) desde WAN, garantizando la comunicación bidireccional necesaria: permitir la entrada de conexiones desde el exterior y que el honeypot pueda responder a dichas solicitudes.

► WAN (tráfico entrante desde exterior)

Se permite el acceso desde WAN al servicio publicado del honeypot únicamente mediante la regla asociada al Port Forward (o su regla equivalente en WAN). De este modo, se limita y controla exactamente qué puerto puede entrar desde el exterior, evitando la exposición del resto de redes internas

Firewall / Rules / WAN

Floating **WAN** LAN DMZ DMZ2

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	0/0 B	IPv4	*	*	192.168.200.99	22 (SSH)	*	none	SSH	
☐	✓	0/0 B	IPv4	TCP	*	*	192.168.200.99	222	*	NAT Reglas honeypot ssh222	

Add Add Delete Toggle Copy Save Separator

NAT (Port Forward)

Se configura un Port Forward desde WAN hacia el honeypot ubicado en DMZ (por ejemplo, TCP 222 → 192.168.200.99:222). Con ello se habilita un acceso externo controlado al servicio publicado, sin necesidad de exponer la DMZ en su conjunto.

Firewall / NAT / Port Forward

Port Forward 1:1 Outbound NPT

Rules												
			Interface	Protocol	Source Address	Source Ports	Dest. Address	Dest. Ports	NAT IP	NAT Ports	Description	Actions
☐	☒		WAN	TCP	*	*	WAN address	222	192.168.200.99	222	Reglas honeypot ssh222	  
☐	☒		WAN	TCP	*	*	WAN address	80 (HTTP)	192.168.200.99	80 (HTTP)	Servidor web Apache	  

Add Add Delete Toggle Save Separator

10

LAN (aislamiento LAN → DMZ)

Se bloquea explícitamente el tráfico desde la LAN hacia el honeypot situado en DMZ (por ejemplo: Block: LAN net 192.168.100.0/24 → 192.168.200.99, en los puertos necesarios o any). Para que el aislamiento sea efectivo, esta regla debe colocarse por encima de cualquier permiso general del tipo “allow LAN to any”.

Firewall / Rules / LAN

Floating WAN LAN DMZ DMZ2

Rules (Drag to Change Order)											
	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒		0/0 B	*	*	LAN Address	443 80	*	*		Anti-Lockout Rule	
☐		0/0 B	IPv4 *	192.168.100.0/24	*	192.168.200.99	*	*	none	Block Lan ->Honeypot	   
☐		27/91.09 MIB	IPv4 *	LAN subnets	*	*	*	*	none	Default allow LAN to any rule	    
☐		0/0 B	IPv6 *	LAN subnets	*	*	*	*	none	Default allow LAN IPv6 to any rule	    

Add Add Delete Toggle Copy Save Separator

DMZ (salida del honeypot hacia redes internas)

Se bloquea la salida del honeypot hacia LAN y DMZ2:

- Block: 192.168.200.99 → LAN net (192.168.100.0/24)
- Block: 192.168.200.99 → DMZ2 net (192.168.250.0/24)

Rules (Drag to Change Order)	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✗ 0/0 B	IPv4 *	192.168.200.99	*	192.168.100.0/24	*	*	none		Block honey -> Lan	
☐	✗ 0/0 B	IPv4 *	192.168.200.99	*	192.168.250.0/24	*	*	none		Block Honey -> DMZ2	
☐	✓ 0/0 B	IPv4 ICMP any	*	*	*	*	*	none		Reglas Trafico ICMP	
☐	✓ 8/838 KiB	IPv4 UDP	*	*	*	53 (DNS)	*	none		regla DNS	
☐	✓ 13/42.97 MiB	IPv4 TCP	*	*	*	webs	*	none		Reglas trafico web	

Buttons: Add, Add, Delete, Toggle, Copy, Save, Separator

DMZ2 (aislamiento DMZ2 → DMZ)

Se bloquea el acceso desde DMZ2 hacia el honeypot ubicado en DMZ, aplicando una regla del tipo Block: DMZ2 net (192.168.250.0/24) → 192.168.200.99. Con ello se mantiene la separación entre el sensor y el honeypot, evitando comunicaciones directas entre ambos segmentos.

Rules (Drag to Change Order)	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✗ 0/0 B	IPv4 *	192.168.250.0/24	*	192.168.200.99	*	*	none		Block DMZ2-> Honeypot	
☐	✓ 0/0 B	IPv4 ICMP any	*	*	*	*	*	none		Reglas Trafico ICMP	
☐	✓ 14/216 KiB	IPv4 UDP	*	*	*	53 (DNS)	*	none		regla DNS	
☐	✓ 22/5.00 MiB	IPv4 TCP	*	*	*	webs	*	none		Regla trafico web	

Buttons: Add, Add, Delete, Toggle, Copy, Save, Separator

Con este conjunto de reglas se aplica una segmentación estricta y una exposición mínima: el honeypot solo queda accesible desde WAN mediante la publicación controlada del servicio (NAT/Port Forward), mientras que el tráfico entre redes internas se restringe según el principio de mínimo privilegio.

3. Políticas e integraciones por agente en el SIEM

Evidencia 1 → Lista de agentes y política asignada

En Elastic Fleet se verifica que cada máquina tiene su Elastic Agent enrolado y que se le ha asignado una política acorde a su función: Windows (LAN) cuenta con una política con integraciones de eventos del sistema (Security/System, según configuración), el honeypot (DMZ) dispone de una política para recoger logs de Cowrie (por ejemplo, vía Filestream sobre los ficheros de Cowrie) y Suricata (DMZ2) tiene una política con la integración de Suricata para la ingestión del eve.json. Cada política define qué logs se recogen y desde qué host, permitiendo distinguir claramente el origen por red.

Fleet							
Centralized management for Elastic Agents.							
Agents Agent policies Enrollment tokens Uninstall tokens Data streams Settings							
Ingest Overview Metrics Agent Info Metrics Agent activity Add agent							
Filter your data using KQL syntax							
Status 3 Tags 1 Agent policy 3 Upgrade available							
Showing 4 agents 3 Healthy 0 Unhealthy 0 Orphaned 0 Updating 0 Offline 1 Inactive 0 Unenrolled 0 Uninstalled 0							
Status	Host	Agent policy	CPU	Memory	Last activity	Version	Actions
Healthy	Windows10	Windows rev. 2	N/A	N/A	20 seconds ago	9.2.4	
Healthy	kali	Honeypot rev. 3	2.29 %	276 MB	5 seconds ago	9.2.4	
Offline	kali	Honeypot rev. 2 Outdated policy	N/A	N/A	19 hours ago	9.2.4	
Healthy	kali	Politica linux Suricata rev. 2	2.77 %	224 MB	10 seconds ago	9.2.4	
Rows per page: 20 < 1 >							

Evidencia 2 → Detalle de cada agente (Policy + Integrations)

[View all agents](#)

kali

[Agent details](#)[Logs](#)[Diagnostics](#)[Settings](#)

Overview

CPU ⓘ3.09 %

Memory ⓘ276 MB

StatusHealthy

Last activityJan 31, 2026 2:55 PM

Last checkin messageRunning

Agent IDacc5b212-3587-4e18-8af3-dca1a6dc91d4

Agent policyHoneypot rev. 3

Agent version9.2.4

Host namekali

Host ID686fa41363f049e9a84be345a7078e04

Output for integrationsDefault output

Output for monitoringDefault output

Logging levelinfo

Privilege modeRunning as root

Agent releasestable

Platformkali

Monitor logsEnabled

Monitor metricsEnabled

Tags-

FIPS modeNot enabled

View more agent metrics

Overview

CPU ⓘ	2.73 %	View more agent metrics
Memory ⓘ	224 MB	
Status	Healthy	
Last activity	Jan 31, 2026 2:56 PM	
Last checkin message	Running	
Agent ID	4aaaa471-65ed-4d8e-ac59-5a5060448d02	
Agent policy	Politica linux Suricata rev. 2	
Agent version	9.2.4	
Host name	kali	
Host ID	686fa41363f049e9a84be345a7078e04	
Output for integrations	Default output	
Output for monitoring	Default output	
Logging level	info	
Privilege mode	Running as root	
Agent release	stable	
Platform	kali	
Monitor logs	Enabled	
Monitor metrics	Enabled	
Tags	-	
FIPS mode	Not enabled	

Integrations

- >  system-1
- > [suricata-1](#)

[View all agents](#)

Windows10

[Agent details](#) [Logs](#) [Diagnostics](#) [Settings](#)

Overview

CPU ⓘ	N/A ⓘ	View more agent metrics
Memory ⓘ	N/A ⓘ	
Status	Healthy	
Last activity	Jan 31, 2026 2:49 PM	
Last checkin message	Running	
Agent ID	fe338c93-e1c4-4b4c-ab20-532937978053	
Agent policy	Windows rev. 2	
Agent version	9.2.4	
Host name	Windows10	
Host ID	b03073f0-8c78-41e8-81ed-eb67037c01c4	
Output for integrations	Default output	
Output for monitoring	Default output	
Logging level	info	
Privilege mode	Running as root	
Agent release	stable	
Platform	windows	
Monitor logs	Enabled	
Monitor metrics	Enabled	
Tags	-	
FIPS mode	Not enabled	

Integrations

- >  system-2
- >  windows-1

[< Cancel](#)



Edit Windows integration

Modify integration settings and deploy changes to the selected agent policies.

1

Configure integration

Integration settings

Choose a name and description to help identify how this integration will be used.

Integration name

windows-1

Description

Optional

[Advanced options](#)

☒ Collect events from the following Windows event log channels:

[Change defaults](#) [^](#)

☒ AppLocker/EXE and DLL

Microsoft-Windows-AppLocker/EXE and DLL channel

Preserve original event

☒

Preserves a raw copy of the original XML event, added to the field event.original

[Advanced options](#)

☒ AppLocker/MSI and Script

Microsoft-Windows-AppLocker/MSI and Script channel

Preserve original event

☒

Preserves a raw copy of the original XML event, added to the field event.original

[Advanced options](#)

☒ Packaged app-Deployment

Microsoft-Windows-AppLocker/Packaged app-Deployment channel

Preserve original event

☒

Preserves a raw copy of the original XML event, added to the field event.original

[Advanced options](#)

[< Cancel](#)



Edit Custom Logs (Filestream) integration

Modify integration settings and deploy changes to the selected agent policies.

1

Configure integration

Integration settings

Choose a name and description to help identify how this integration will be used.

Integration name

Honeypot

Description

Optional

[Advanced options](#)

☒ Custom Filestream Logs

[Change defaults](#) [v](#)

Custom Filestream Logs

2

Where to add this integration?

For existing hosts:

Agent policies

Agent policies are used to manage a group of integrations across a set of agents.

Agent policies

Honeypot [x](#) [v](#)

2 agents are enrolled with the selected agent policies.

For a new host:

[+ Create a new agent policy](#)

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[< Cancel](#)

Edit Suricata integration

Modify integration settings and deploy changes to the selected agent policies.

1

Configure integration

Integration settings

Choose a name and description to help identify how this integration will be used.

Integration name

suricata-1

Description

Optional

[> Advanced options](#)

☒ **Collect Suricata eve logs (input: logfile)**

[Change defaults ^](#)

Collect Suricata eve logs using log input

Paths

/var/log/suricata/eve.json

[⊕ Add row](#)

Preserve original event

☐ 

Preserves a raw copy of the original event, added to the field event.original

[> Advanced options](#)

2

Where to add this integration?

For existing hosts:

En cada política se revisa la sección 'Integrations', donde se ve qué tipo de logs se recogen en cada máquina.

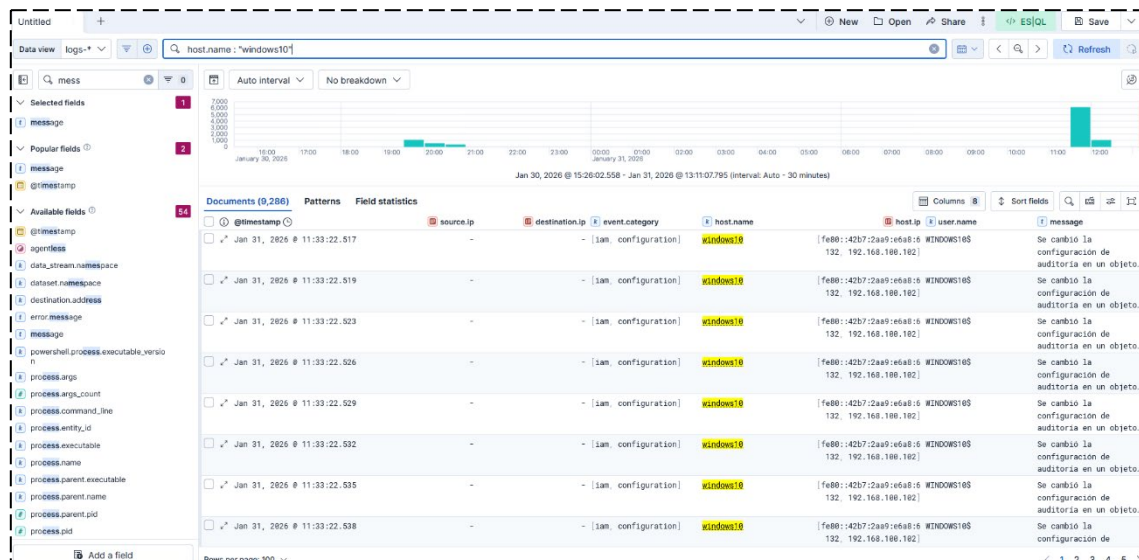
4. Evidencias de correcta recepción de logs en el SIEM (Elastic) y esquema por redes

Se comprueba que el SIEM recibe logs de las tres fuentes requeridas y que cada una corresponde a su red: Windows genera eventos desde LAN (192.168.100.0/24), Cowrie genera eventos desde DMZ (192.168.200.0/24) y Suricata genera eventos desde DMZ2 (192.168.250.0/24). La validación se realiza revisando campos típicos como host.ip, agent.name, source.ip, destination.ip y el dataset correspondiente, confirmando que cada evento queda asociado a la máquina y a la red correctas.

17

➤ **Logs de Windows (red LAN)**

Demuestra que el equipo Windows pertenece a la LAN y que está enviando correctamente eventos a Elastic.



➤ Logs del Honeypot Cowrie (red DMZ)

Demuestra que el honeypot Cowrie está ubicado en la DMZ y que está enviando correctamente sus logs a Elastic.

Documents (126)			Patterns	Field statistics	Columns 8	Sort fields	Q	+	-	⌵
⌵ @timestamp	source.ip	destination.ip	event.category	host.name	host.ip	user.name	message			
✓ Jan 30, 2026 @ 17:00:32.264	-	- (null)	kali	[192.168.200.99, (null)]	fe80::1928:c05a:bc3d:c364, 172.17.0...	(null)	2026-01-30T15:28:25+00:00 0 [-] Reading configuration from ['/'			
✓ Jan 30, 2026 @ 17:00:34.183	-	- (null)	kali	[192.168.200.99, (null)]	fe80::1928:c05a:bc3d:c364, 172.17.0...	(null)	2026-01-30T15:28:25+00:00 0 [twisted.scripts._twis			
✓ Jan 30, 2026 @ 17:00:34.183	-	- (null)	kali	[192.168.200.99, (null)]	fe80::1928:c05a:bc3d:c364, 172.17.0...	(null)	2026-01-30T15:28:25+00:00 0 [-] Cowrie Version 2.9.9.dev+q88b65ffa6			
✓ Jan 30, 2026 @ 19:14:01.952	-	- (null)	kali	[192.168.200.99, (null)]	fe80::1928:c05a:bc3d:c364, 172.17.0...	(null)	2026-01-30T18:10:58+00:00 0 [twisted.scripts._twis			
✓ Jan 30, 2026 @ 19:14:01.952	-	- (null)	kali	[192.168.200.99, (null)]	fe80::1928:c05a:bc3d:c364, 172.17.0...	(null)	2026-01-30T18:10:58+00:00 0 [-] Cowrie Version 2.9.9.dev+q88b65ffa6			
✓ Jan 30, 2026 @ 19:14:01.952	-	- (null)	kali	[192.168.200.99, (null)]	fe80::1928:c05a:bc3d:c364, 172.17.0...	(null)	2026-01-30T18:10:58+00:00 0 [-] Reading configuration from ['/'			
✓ Jan 30, 2026 @ 19:14:01.953	-	- (null)	kali	[192.168.200.99, (null)]	fe80::1928:c05a:bc3d:c364, 172.17.0...	(null)	2026-01-30T18:13:59+00:00 0 [honeypot.ssh.transport			
✓ Jan 30, 2026 @ 19:14:01.953	-	- (null)	kali	[192.168.200.99, (null)]	fe80::1928:c05a:bc3d:c364, 172.17.0...	(null)	2026-01-30T18:13:59+00:00 0			

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➤ Logs de Suricata (red DMZ2)

Demuestra que Suricata está ubicado en la DMZ2 y que está enviando correctamente sus eventos a Elastic.

My Security project

Discover

Security

Discover

Dashboards

Rules

Alerts

Attack discovery

Findings

Cases

More

Untitled

+

▼

New

Open

Share

ES/SQL

Save

Data view

suricata

event.dataset:"suricata eve" and suricata eve.event.type:"alert" and suricata eve.alert.signature:"tráfico detectado"

Jan 30, 2026 @ 15:2...

now

Refresh

event

Auto Interval

No breakdown

800

600

400

200

0

0

10:00

11:00

12:00

13:00

14:00

15:00

16:00

17:00

18:00

19:00

20:00

21:00

22:00

23:00

00:00

01:00

02:00

03:00

04:00

05:00

06:00

07:00

08:00

09:00

10:00

11:00

12:00

13:00

Jan 31, 2026

Jan 30, 2026 @ 15:26:02:556 - Jan 31, 2026 @ 13:12:34:727 Interval: Auto - 30 minutes

Documents (2,686)

Patterns

Field statistics

Columns 9

Sort fields 1

Q

+

-

⌵

⌵ @timestamp

source.ip

destination.ip

event.category

host.name

host.ip

message

suricata eve.alert.sig

network.transport

✓ Jan 31, 2026 @ 13:12:22.167

34.68.177.88

192.168.258.161

[network, intrusion_detection]

(null)

- (blank)

tráfico detectado

tcp

✓ Jan 31, 2026 @ 13:12:22.840

192.168.258.161

34.68.177.88

[network, intrusion_detection]

(null)

- (blank)

tráfico detectado

tcp

✓ Jan 31, 2026 @ 13:12:13.371

34.68.177.88

192.168.258.161

[network, intrusion_detection]

(null)

- (blank)

tráfico detectado

tcp

✓ Jan 31, 2026 @ 13:12:13.244

192.168.258.161

34.68.177.88

[network, intrusion_detection]

(null)

- (blank)

tráfico detectado

tcp

✓ Jan 31, 2026 @ 13:12:12.166

34.68.177.88

192.168.258.161

[network, intrusion_detection]

(null)

- (blank)

tráfico detectado

tcp

✓ Jan 31, 2026 @ 13:12:12.839

192.168.258.161

34.68.177.88

[network, intrusion_detection]

(null)

- (blank)

tráfico detectado

tcp

✓ Jan 31, 2026 @ 13:12:02.366

34.68.177.88

192.168.258.161

[network, intrusion_detection]

(null)

- (blank)

tráfico detectado

tcp

✓ Jan 31, 2026 @ 13:12:02.237

192.168.258.161

34.68.177.88

[network, intrusion_detection]

(null)

- (blank)

tráfico detectado

tcp

Rows per page: 100

1

2

3

4

5

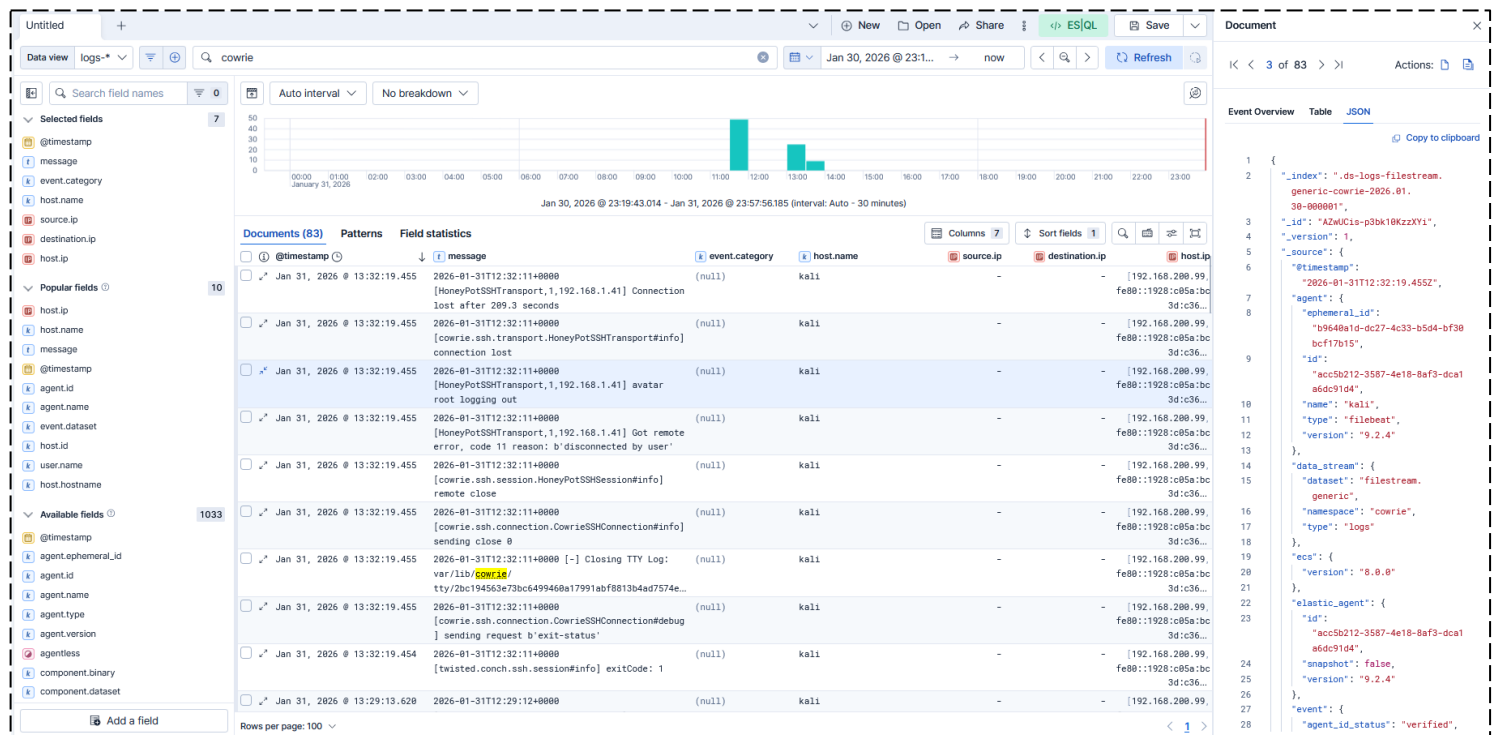
5. Explicación de los logs seleccionados (campos y significado)

Para cada fuente, se interpretan los campos principales usados como prueba:

HoneyPot: Cowrie (DMZ)

La identificación se realiza mediante mensajes característicos del servicio de honeypot y eventos de sesión/comandos. Como campos clave se revisan @timestamp (momento), host.ip (IP del honeypot en DMZ), message (acción/comando) y agent.* (agente que envía).

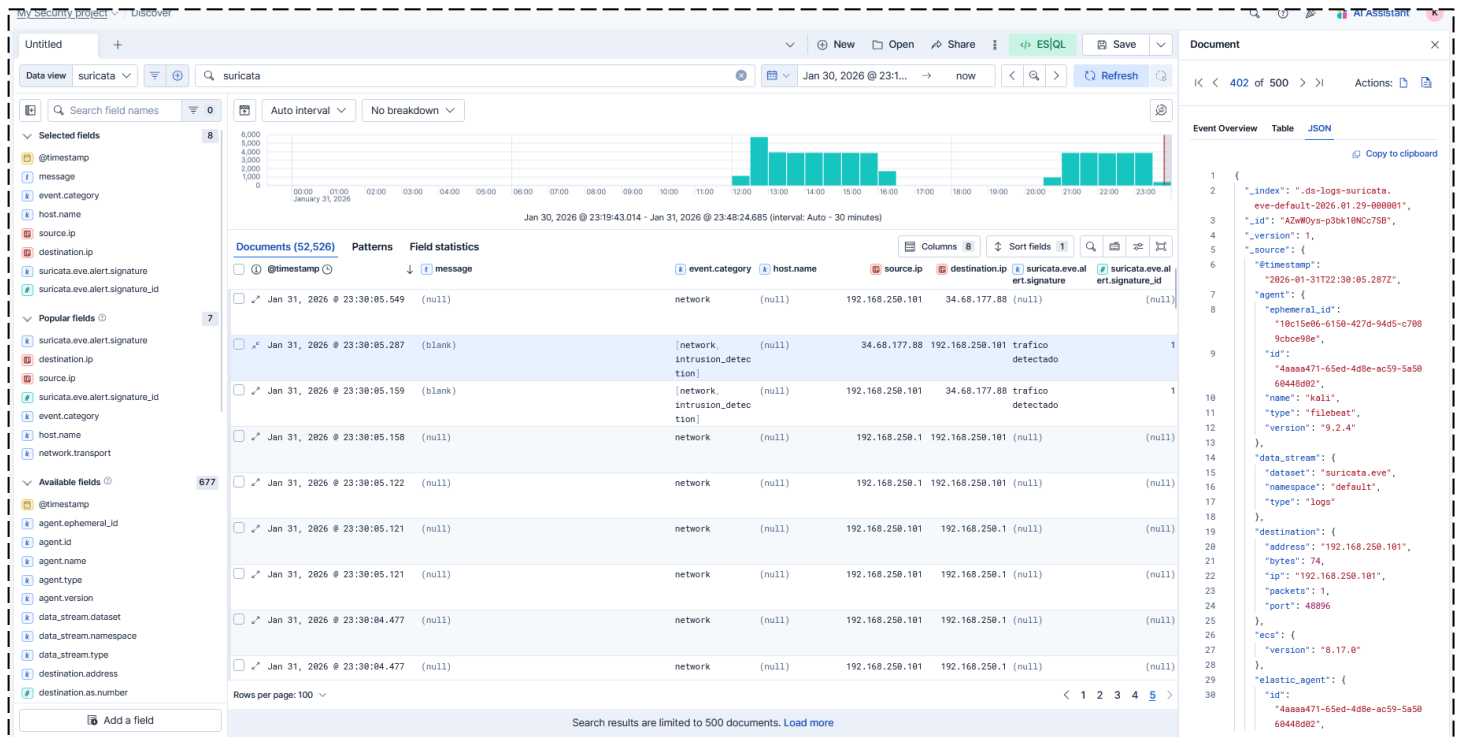
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Suricata (DMZ2)

La identificación se realiza mediante eventos/alertas en el dataset de Suricata (por ejemplo, suricata.eve). Como campos clave se revisan @timestamp, source.ip, destination.ip, event.type/alert.signature (si hay alerta) y host.ip (IP del sensor en DMZ2).

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The screenshot displays the Elastic Security interface for analyzing Suricata events. The top panel shows a bar chart of event counts over time, with a peak around 12:00 on Jan 30, 2026. The middle panel shows a table of documents with columns for @timestamp, message, event.category, host.name, source.ip, destination.ip, suricata.eve.alert.signature, and suricata.eve.alert.signature_id. The bottom panel shows the JSON document structure for a specific event.

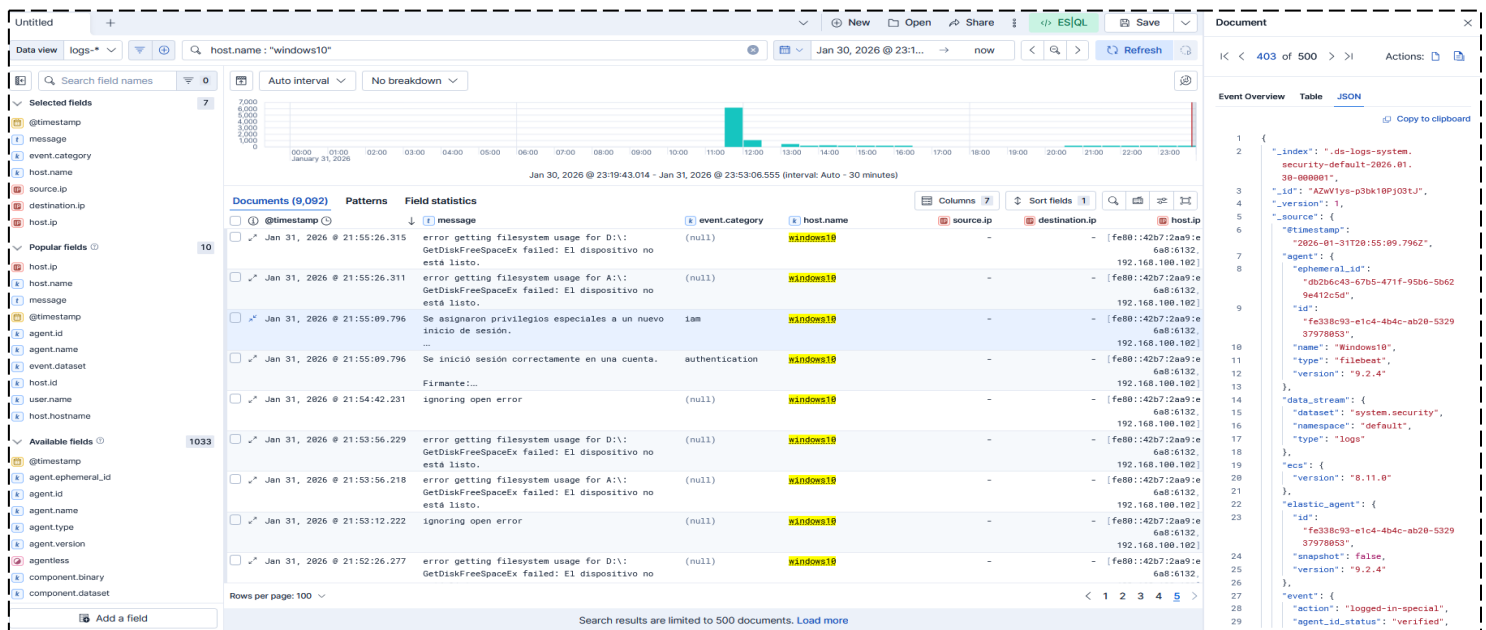
Document	@timestamp	message	event.category	host.name	source.ip	destination.ip	suricata.eve.alert.signature	suricata.eve.alert.signature_id
1	Jan 31, 2026 @ 23:38:05.549	(null)	network	(null)	192.168.250.101	34.68.177.88	(null)	(null)
2	Jan 31, 2026 @ 23:38:05.287	(blank)	[network, intrusion_detection]	(null)	34.68.177.88	192.168.250.101	trafico detectado	1
3	Jan 31, 2026 @ 23:38:05.159	(blank)	[network, intrusion_detection]	(null)	192.168.250.101	34.68.177.88	trafico detectado	1
4	Jan 31, 2026 @ 23:38:05.158	(null)	network	(null)	192.168.250.1	192.168.250.101	(null)	(null)
5	Jan 31, 2026 @ 23:38:05.122	(null)	network	(null)	192.168.250.1	192.168.250.101	(null)	(null)
6	Jan 31, 2026 @ 23:38:05.121	(null)	network	(null)	192.168.250.101	192.168.250.1	(null)	(null)
7	Jan 31, 2026 @ 23:38:05.121	(null)	network	(null)	192.168.250.101	192.168.250.1	(null)	(null)
8	Jan 31, 2026 @ 23:38:04.477	(null)	network	(null)	192.168.250.101	192.168.250.1	(null)	(null)
9	Jan 31, 2026 @ 23:38:04.477	(null)	network	(null)	192.168.250.101	192.168.250.1	(null)	(null)

```
{
  "_index": ".ds-logs-suricata",
  "_type": "suricata.eve",
  "_id": "AZw0Dys-p3bk10NDC75B",
  "_version": 1,
  "_source": {
    "@timestamp": "2026-01-31T22:38:05.287Z",
    "agent": {
      "ephemeral_id": "18c15e86-6150-4270-94d5-c788",
      "id": "9bce99be",
      "name": "kali",
      "type": "filebeat",
      "version": "9.2.4"
    },
    "data_stream": {
      "dataset": "suricata.eve",
      "namespace": "default",
      "type": "logs"
    },
    "destination": {
      "address": "192.168.250.101",
      "bytes": 74,
      "ip": "192.168.250.101",
      "packets": 1,
      "port": 48896
    },
    "ecs": {
      "version": "8.17.0"
    },
    "elastic_agent": {
      "id": "4a8aa471-65ed-4d8e-ac59-5a5068448092"
    }
  }
}
```

Windows (LAN)

La identificación se realiza mediante eventos del sistema/seguridad asociados al host de Windows. Como campos clave se revisan host.name, host.ip (LAN), event.code (ID del evento), message/winlog.* (detalle) y @timestamp.

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Esta actividad permite crear una red segmentada y segura con pfSense (LAN, DMZ y DMZ2), controlar los accesos mediante reglas de firewall y validar la configuración con evidencias. Además, posibilita monitorizar y centralizar los logs en un SIEM para correlacionar eventos, detectar comportamientos sospechosos y mantener trazabilidad de lo que ocurre en la infraestructura. Con el honeypot Cowrie y Suricata se generan eventos reales de ataque y tráfico, que el SIEM recoge y visualiza, facilitando la detección temprana, el análisis y la respuesta ante incidentes.